

HARI RATHNAM S

Aspiring Software Engineer | B.E. Computer Science, SRM Easwari Engineering College (2028)
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PROFESSIONAL SUMMARY

Second-year Computer Science student (CGPA 8.65) with hands-on experience in DSA mentoring, data visualization, and machine learning pipelines. Demonstrated ability to deliver functional prototypes under pressure through national-level hackathons and remote internships. Seeking software engineering opportunities to apply strong problem-solving and full-stack foundations.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript

Libraries & Frameworks: React.js, pandas, NumPy, scikit-learn, Matplotlib, HTML5, CSS3, SQL

Tools & Platforms: Git, GitHub, VS Code, Tableau Public, AWS (Cloud 101, Generative AI), MS Office

CS Fundamentals: Data Structures & Algorithms, OOP, System Design (Basics), Competitive Programming

EXPERIENCE

DSA & LeetCode Peer Tutor | SRM Easwari Engineering College, Chennai

Nov 2025 – Jan 2026

- Appointed by faculty to lead weekly DSA sessions for 20+ first-year students, covering Big-O analysis, dynamic programming, greedy algorithms, and recursion.
- Guided students through 40+ LeetCode problems with structured problem-solving frameworks; provided personalized feedback on time/space complexity optimizations.
- Strengthened technical communication and mentoring skills; consistently received positive peer feedback on clarity and depth of explanations.

Data Science Intern | 1M1B Green Internship (Remote)

Dec 2025 – Jan 2026

- Built a Tourist Carbon Footprint Dashboard in Tableau Public by cleaning and processing real-world datasets across 5+ tourism categories to surface sustainability insights.
- Designed 4 interactive dashboards with dynamic filters and calculated fields, enabling data-driven reporting on carbon emission trends for non-technical stakeholders.
- Applied data wrangling techniques to normalize inconsistent source data, improving dashboard accuracy and reliability for environmental KPI tracking.

PROJECTS

Smart Hospital Queue Management System

Java, DSA, Priority Queue, Load Balancing | 2026

- Engineered a min-heap priority queue to schedule patients by severity, simulating a 40% reduction in average wait time with dynamic load balancing across N doctor queues.
- Implemented multi-doctor workload distribution ensuring no resource exceeded 80% utilization; validated edge cases including emergency triage and doctor unavailability via unit tests.

Urban Crowd Detection & Breakdown Prediction System

Python, Machine Learning, Data Analysis | 2026

- Built an ML classification pipeline using scikit-learn to detect abnormal crowd density in real time and predict congestion events up to 10 minutes ahead, achieving 85%+ F1-score on validation data.
- Applied feature engineering and evaluated models on precision, recall, and F1-score; architected system for scalable real-time data ingestion in smart-city environments.
- Designed the system to process 1,000+ data points per minute, making it viable for deployment in high-footfall public venues and transportation hubs.

EDUCATION

B.E. Computer Science & Engineering — CGPA: 8.65 / 10

SRM Easwari Engineering College, Chennai | 2024 – 2028

(Expected)

HSC (Class XII) — 90% | SSLC (Class X) — 91%

St. Joseph Matriculation Higher Secondary School | 2024 / 2022

CERTIFICATIONS & ACHIEVEMENTS

- AWS: Introduction to Cloud 101 — Amazon Web Services (2026)
- AWS: Introduction to Generative AI — Amazon Web Services (2026)
- Introduction to Industry 4.0 & Industrial Internet of Things — NPTEL (Nov 2025)
- Secured an internship offer through a hackathon competition; delivered a functional prototype within a 48-hour window as part of a cross-functional team.

LANGUAGES

Tamil (C2 — Bilingual Proficient) English (B2 — Upper Intermediate) Hindi (A2 — Elementary)